WASP-10b

R-1511

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-10_131025` · created 2026-08-01T15:59:19+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456590.8076 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.012
Transit depth (Rp/Rs)^2	2.31 +/- 0.38 [%]
Orbital Inclination (inc)	89.73 +/- 0.63
Airmass coefficient 1 (a1)	0.99 +/- 0.017
Airmass coefficient 2 (a2)	0.0094 +/- 0.0071
Scatter in the residuals of the lightcurve fit is	1.7 %
Best Comparison Star	#1 - [334, 355]
Optimal Aperture	3.47
Optimal Annulus	10.31
Transit Duration (day)	0.0941 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.012 vs 0.1592 (deviation 0.55σ)
Duration fit vs published	0.0941 d vs 0.0946 d (deviation 0.32σ)
Mid-time O–C	-0.3 min
Depth SNR	11.6
Residual scatter	1.7 %

Light curve

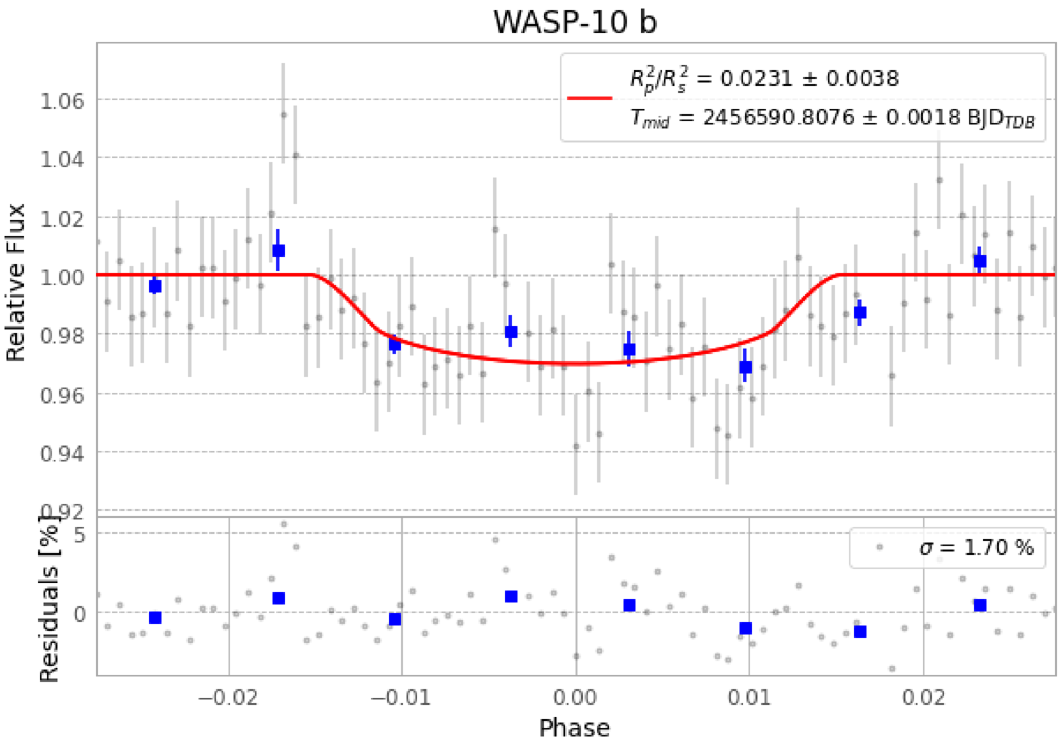

Figure 1 — FinalLightCurve_WASP-10 b_2013-10-24.png (SHA-256 f2e449ebb49c786c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 283], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 e25cf9a2ddad91b1…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

83 FITS frames (55 MB), `WASP-10131025051212.FITS` → `WASP-10131025091812.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-131025.log (aee1eca0e472…), FinalLightCurve_WASP-10 b_2013-10-24.png (f2e449ebb49c…), FinalLightCurve_WASP-10 b_2013-10-24.csv (296a1d9b7d89…)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 15:59Z.